



**NATIONAL
SENIOR CERTIFICATE**

GRADE 11

NOVEMBER 2022

**PHYSICAL SCIENCES P2
(CHEMISTRY)**

MARKS: 100

TIME: 2 hours

This question paper consists of 16 pages, including 4 data sheets.

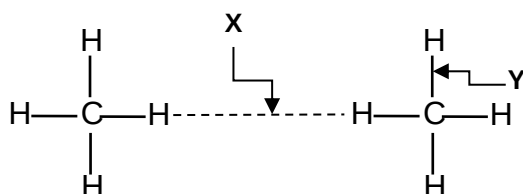
INSTRUCTIONS AND INFORMATION

1. Write your full NAME and SURNAME in the appropriate spaces on the ANSWER BOOK.
2. This question paper consists of SEVEN questions. Answer ALL the questions in the ANSWER BOOK.
3. Start EACH question on a NEW page in the ANSWER BOOK.
4. Number the answers correctly according to the numbering system used in this question paper.
5. Leave ONE line between two sub-questions, for example between QUESTION 2.1 and QUESTION 2.2.
6. You may use a non-programmable calculator.
7. You may use appropriate mathematical instruments.
8. Show ALL formulae and substitutions in ALL calculations.
9. Round off your FINAL numerical answers to a minimum of TWO decimal places.
10. Give brief motivations, discussions, et cetera where required.
11. You are advised to use the attached DATA SHEETS.
12. Write neatly and legibly.

QUESTION 1: MULTIPLE-CHOICE QUESTIONS

Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question numbers (1.1 to 1.7) in the ANSWER BOOK, for example 1.8 D.

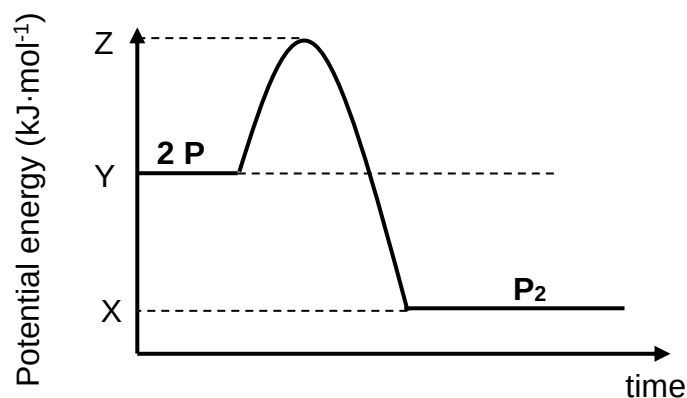
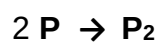
- 1.1 Which ONE of the following is the NAME given to the reaction in which electrons are transferred from one substance to another substance?
- A Redox
- B Precipitation
- C Neutralisation
- D Decomposition (2)
- 1.2 The diagram below is used to demonstrate the differences between interatomic bonds (chemical bonds) and intermolecular forces.



Which ONE of the following statements is CORRECT?

- A X is stronger than Y
- B X is an interatomic bond
- C X is an intermolecular force
- D Y is an intermolecular force (2)
- 1.3 The minimum amount of energy needed for a chemical reaction to start is called ...
- A activation energy.
- B catalyst.
- C heat of reaction.
- D bond energy. (2)

1.4 Consider the potential energy graph for the hypothetical reaction:



The heat of the reaction (ΔH) is represented by ...

A $Z - Y$.

B $X - Y$.

C $Y - X$.

D $Y - Z$.

(2)

1.5 Consider the INCOMPLETE reaction below:



Which ONE of the following is CORRECT regarding compound X?

Compound X is a(n) ...

A carbonate.

B metal.

C metal oxide.

D acid.

(2)

1.6 Consider the substances **P**, **Q** and **R**.

P dissolves in **Q** but not in **R**.

The MOST LIKELY bonds or intermolecular forces in substances **P**, **Q** and **R** are:

	P	Q	R
A	Ionic bonds	Dipole-dipole forces	London forces
B	London forces	Dipole-dipole forces	Ionic bonds
C	Dipole-dipole forces	London forces	Ionic bonds
D	Ionic bonds	London forces	Dipole-dipole forces

(2)

1.7 According to the kinetic molecular theory, molecules of gases at the **same temperature** always have the same ...

- A mass.
- B volume.
- C pressure.
- D average kinetic energy.

(2)

[14]

QUESTION 2 (Start on a new page.)

Consider the following compounds.



2.1 Define the term *molecule*. (2)

2.2 Draw the Lewis structures for:

2.2.1 O_2 (2)

2.2.2 H_2O (2)

2.3 The H_3O^+ - ions forms when the oxygen atom in H_2O donates its lone pair of electrons into the vacant orbital of H^+ .

2.3.1 Write down the NAME of the type of bond described by the underlined phrase. (1)

2.3.2 Draw the Lewis structure for the H_3O^+ -ion. (2)

2.4 Which molecule is polar, H_2O or O_2 ?

Explain your answer. (4)

[13]

QUESTION 3 (Start on a new page.)

The table below shows the relationship between the melting points of three molecules:

Molecule	Melting point (°C)
CH ₄	-182,5
CF ₄	-150
CCl ₄	-23

- 3.1 Define the term *melting point*. (2)
- 3.2 Explain the trend in melting points of the molecules in the above table by referring to the intermolecular forces and energy involved. (4)
- 3.3 Which molecule in the table will have the highest vapour pressure at a given temperature?
Explain the answer by referring to the data in the table. (2)
- 3.4 Water (H₂O) is a smaller molecule than CCl₄, but water has a higher melting point than CCl₄.
Explain this observation by referring to the type of intermolecular forces involved. (2)
- 3.5 Write down the name of intermolecular force that will exist in a mixture of H₂O and CCl₄. (2)

[12]

QUESTION 4 (Start on a new page.)

A group of learners investigate the relationship between pressure and the volume of an enclosed gas at room temperature.

They recorded their results in the table below:

Pressure (kPa)	Volume (cm ³)
100,33	7,34
102,2	7,21
103,93	7,09
X	6,97

- 4.1 For this investigation, write down the:
- 4.1.1 Name of the gas law that is being investigated (1)
- 4.1.2 Controlled variable (1)
- 4.1.3 Relationship between pressure and volume of the gas as described by the first 3 data sets in the table (2)
- 4.2 Calculate the value of X. (4)
- 4.3 Write down the TWO conditions under which real gases behave more like an ideal gas. (2)
- 4.4. Write down the NAME of a gas whose behaviour is close to that of an ideal gas, under the conditions mentioned in QUESTION 4.3 above. (1)
- [11]**

QUESTION 5 (Start on a new page.)

5.1 An organic compound has the following composition by mass.

Element	Percentage (%)
C	54,55
H	9,09
O	36,36

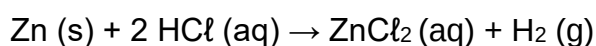
The molar mass of the compound is TWICE the empirical molar mass.

5.1.1 Define the term *empirical formula*. (2)

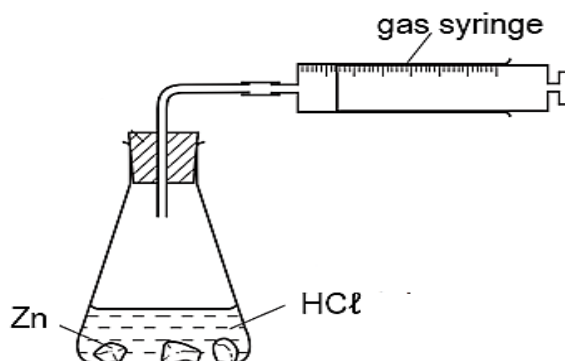
5.1.2 Determine, by calculation, the molecular formula of the compound. (5)

5.2 A teacher demonstrates to a group of learners the impact that the limiting reagent have on the amount of product formed during a chemical reaction.

The teacher uses the following chemical equation:



The teacher uses the following set-up for this investigation:



Experiment 1	Experiment 2
7 g of Zn	3,27 g of Zn
1 mol of HCl	1 mol of HCl

5.2.1 Define the term *limiting reagent*. (2)

5.2.2 Determine, by calculation, the limiting reagent in **experiment 1**. (5)

- 5.2.3 How will the amount of zinc chloride (ZnCl_2) produced in **experiment 2**, compare to that of **experiment 1**?

Write down only HIGHER THAN, LOWER THAN or EQUAL TO.

Explain your answer.

(4)

- 5.2.4 **Experiment 2** was carried out at 40°C . The molar volume of hydrogen gas at this temperature is $25,7 \text{ dm}^3 \cdot \text{mol}^{-1}$.

Calculate the volume of hydrogen gas produced once the reaction reaches completion in **experiment 2**.

(5)

[23]

QUESTION 6 (Start on a new page.)

6.1 Consider the following ionisation reaction of HCl:

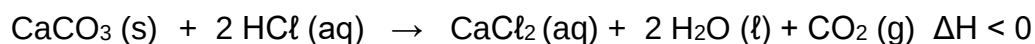


6.1.1 Define an *acid* in terms of the Lowry-Brønsted model. (2)

6.1.2 Write down ONE conjugate acid-base pair. (2)

6.1.3 Write down the formula of the substance in the reaction that can act as an ampholyte in some reactions. (1)

6.2 A certain volume of a hydrochloric acid solution of concentration $0,5 \text{ mol.dm}^{-3}$ is added to a sample of seashells of mass 8 g in a container. The hydrochloric acid solution (HCl) COMPLETELY reacts with the calcium carbonate (CaCO_3) in the sample according to the balanced equation:



The seashell contains 95% CaCO_3 .

6.2.1 Is the reaction ENDOTHERMIC or EXOTHERMIC?

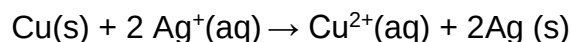
Give a reason for the answer. (2)

6.2.2 Calculate the volume, in cm^3 , of the hydrochloric acid (HCl) that was added to the sample inside the container. (7)

[14]

QUESTION 7 (Start on a new page.)

7.1 Consider the balanced redox reaction:



7.1.1 Define *oxidation*. (2)

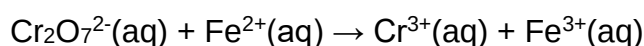
From the balanced equation above, write down the:

7.1.2 Reduction half reaction (2)

7.1.3 Formula of the reducing agent (1)

7.1.4 Explanation for the answer in QUESTION 7.1.3 by referring to the oxidation numbers (2)

7.2 The following INCOMPLETE chemical equation represents a redox reaction:



7.2.1 Determine the oxidation number of Cr in $\text{Cr}_2\text{O}_7^{2-}$. (2)

7.2.2 Using information from the table of standard reduction potentials write down the BALANCED net ionic reaction.

Clearly indicate the following:

- Reduction half reaction
 - Oxidation half reaction
 - Balanced net ionic equation
- (4)
[13]

TOTAL: 100

**NATIONAL SENIOR CERTIFICATE
NASIONALE SENIOR SERTIFIKAAT**

**DATA FOR PHYSICAL SCIENCES GRADE 12
PAPER 2 (CHEMISTRY)**

**GEGEWENS VIR FISIESTE WETENSKAPPE GRAAD 12
VRAESTEL 2 (CHEMIE)**

TABLE 1: PHYSICAL CONSTANTS/TABEL 1: FISIESTE KONSTANTES

NAAM/NAME	SIMBOOL/SYMBOL	WAARDE/VALUE
Standard pressure <i>Standaarddruk</i>	p^θ	$1,013 \times 10^5 \text{ Pa}$
Molar gas volume at STP <i>Molêre gasvolume teen STD</i>	V_m	$22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$
Standard temperature <i>Standaardtemperatuur</i>	T^θ	273 K
Charge on electron <i>Lading op elektron</i>	e	$-1,6 \times 10^{-19} \text{ C}$
Avogadro's constant <i>Avogadro se konstante</i>	N_A	$6,02 \times 10^{23} \text{ mol}^{-1}$

TABLE 2: FORMULAE/TABEL 2: FORMULES

$n = \frac{m}{M} \text{ or/of}$ $n = \frac{N}{N_A} \text{ or/of}$ $n = \frac{V}{V_m}$	$c = \frac{n}{V} \text{ or/of } c = \frac{m}{MV}$ $\frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}$	$\text{pH} = -\log[\text{H}_3\text{O}^+]$ $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1 \times 10^{-14}$ at /by 298K
$E^\theta_{\text{cell}} = E^\theta_{\text{cathode}} - E^\theta_{\text{anode}} / E^\theta_{\text{sel}} = E^\theta_{\text{katode}} - E^\theta_{\text{anode}}$ $E^\theta_{\text{cell}} = E^\theta_{\text{reduction}} - E^\theta_{\text{oxidation}} / E^\theta_{\text{sel}} = E^\theta_{\text{reduksie}} - E^\theta_{\text{oksidasie}}$ $E^\theta_{\text{cell}} = E^\theta_{\text{oxidising agent}} - E^\theta_{\text{reducing agent}} / E^\theta_{\text{sel}} = E^\theta_{\text{oksideermiddel}} - E^\theta_{\text{reduseermiddel}}$		

TABLE 3: THE PERIODIC TABLE OF ELEMENTS/TABEL 3: DIE PERIODIEKE TABEL VAN ELEMENTE

1 (I)	2 (II)	3	4	5	6	7	8	9	10	11	12	13 (III)	14 (IV)	15 (V)	16 (VI)	17 (VII)	18 (VIII)
KEY/ SLEUTEL																	
Atoomgetal Atomic number																	
Elektronegatiwiteit Electronegativity																	
Simbool Symbol																	
Benaderde relatiewe atoommassa Approximate relative atomic mass																	
1 1 H								29 19 Cu 63,5									2 He 4
3 3 Li 7	4 4 Be 9											5 5 B 11	6 6 C 12	7 7 N 14	8 8 O 16	9 9 F 19	10 Ne 20
11 11 Na 23	12 12 Mg 24											13 13 Al 27	14 14 Si 28	15 15 P 31	16 16 S 32	17 17 Cl 35,5	18 Ar 40
19 19 K 39	20 20 Ca 40	21 21 Sc 45	22 22 Ti 48	23 23 V 51	24 24 Cr 52	25 25 Mn 55	26 26 Fe 56	27 27 Co 59	28 28 Ni 59	29 29 Cu 63,5	30 30 Zn 65	31 31 Ga 70	32 32 Ge 73	33 33 As 75	34 34 Se 79	35 35 Br 80	36 Kr 84
37 37 Rb 86	38 38 Sr 88	39 39 Y 89	40 40 Zr 91	41 41 Nb 92	42 42 Mo 96	43 43 Tc 98	44 44 Ru 101	45 45 Rh 103	46 46 Pd 106	47 47 Ag 108	48 48 Cd 112	49 49 In 115	50 50 Sn 119	51 51 Sb 122	52 52 Te 128	53 53 I 127	54 Xe 131
55 55 Cs 133	56 56 Ba 137	57 57 La 139	72 72 Hf 179	73 73 Ta 181	74 74 W 184	75 75 Re 186	76 76 Os 190	77 77 Ir 192	78 78 Pt 195	79 79 Au 197	80 80 Hg 201	81 81 Tl 204	82 82 Pb 207	83 83 Bi 209	84 84 Po 209	85 85 At 210	86 Rn
87 87 Fr 226	88 88 Ra 226	89 Ac															
			58 Ce 140	59 Pr 141	60 Nd 144	61 Pm	62 Sm 150	63 Eu 152	64 Gd 157	65 Tb 159	66 Dy 163	67 Ho 165	68 Er 167	69 Tm 169	70 Yb 173	71 Lu 175	
			90 Th 232	91 Pa	92 U 238	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

TABLE 4A: STANDARD REDUCTION POTENTIALS
TABEL 4A: STANDAARD REDUKSIEPOTENSIALE

Half-reactions/Halfreaksies	E^{θ} (V)
$F_2(g) + 2e^- \rightleftharpoons 2F^-$	+ 2,87
$Co^{3+} + e^- \rightleftharpoons Co^{2+}$	+ 1,81
$H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O$	+1,77
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	+ 1,51
$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-$	+ 1,36
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+ 1,33
$O_2(g) + 4H^+ + 4e^- \rightleftharpoons 2H_2O$	+ 1,23
$MnO_2 + 4H^+ + 2e^- \rightleftharpoons Mn^{2+} + 2H_2O$	+ 1,23
$Pt^{2+} + 2e^- \rightleftharpoons Pt$	+ 1,20
$Br_2(l) + 2e^- \rightleftharpoons 2Br^-$	+ 1,07
$NO_3^- + 4H^+ + 3e^- \rightleftharpoons NO(g) + 2H_2O$	+ 0,96
$Hg^{2+} + 2e^- \rightleftharpoons Hg(l)$	+ 0,85
$Ag^+ + e^- \rightleftharpoons Ag$	+ 0,80
$NO_3^- + 2H^+ + e^- \rightleftharpoons NO_2(g) + H_2O$	+ 0,80
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+ 0,77
$O_2(g) + 2H^+ + 2e^- \rightleftharpoons H_2O_2$	+ 0,68
$I_2 + 2e^- \rightleftharpoons 2I^-$	+ 0,54
$Cu^+ + e^- \rightleftharpoons Cu$	+ 0,52
$SO_2 + 4H^+ + 4e^- \rightleftharpoons S + 2H_2O$	+ 0,45
$2H_2O + O_2 + 4e^- \rightleftharpoons 4OH^-$	+ 0,40
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+ 0,34
$SO_4^{2-} + 4H^+ + 2e^- \rightleftharpoons SO_2(g) + 2H_2O$	+ 0,17
$Cu^{2+} + e^- \rightleftharpoons Cu^+$	+ 0,16
$Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}$	+ 0,15
$S + 2H^+ + 2e^- \rightleftharpoons H_2S(g)$	+ 0,14
$2H^+ + 2e^- \rightleftharpoons H_2(g)$	0,00
$Fe^{3+} + 3e^- \rightleftharpoons Fe$	- 0,06
$Pb^{2+} + 2e^- \rightleftharpoons Pb$	- 0,13
$Sn^{2+} + 2e^- \rightleftharpoons Sn$	- 0,14
$Ni^{2+} + 2e^- \rightleftharpoons Ni$	- 0,27
$Co^{2+} + 2e^- \rightleftharpoons Co$	- 0,28
$Cd^{2+} + 2e^- \rightleftharpoons Cd$	- 0,40
$Cr^{3+} + e^- \rightleftharpoons Cr^{2+}$	- 0,41
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	- 0,44
$Cr^{3+} + 3e^- \rightleftharpoons Cr$	- 0,74
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	- 0,76
$2H_2O + 2e^- \rightleftharpoons H_2(g) + 2OH^-$	- 0,83
$Cr^{2+} + 2e^- \rightleftharpoons Cr$	- 0,91
$Mn^{2+} + 2e^- \rightleftharpoons Mn$	- 1,18
$Al^{3+} + 3e^- \rightleftharpoons Al$	- 1,66
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	- 2,36
$Na^+ + e^- \rightleftharpoons Na$	- 2,71
$Ca^{2+} + 2e^- \rightleftharpoons Ca$	- 2,87
$Sr^{2+} + 2e^- \rightleftharpoons Sr$	- 2,89
$Ba^{2+} + 2e^- \rightleftharpoons Ba$	- 2,90
$Cs^+ + e^- \rightleftharpoons Cs$	- 2,92
$K^+ + e^- \rightleftharpoons K$	- 2,93
$Li^+ + e^- \rightleftharpoons Li$	- 3,05

Increasing oxidising ability/Toenemende oksiderende vermoë

Increasing reducing ability/Toenemende reduserende vermoë

TABLE 4B: STANDARD REDUCTION POTENTIALS
TABEL 4B: STANDAARD REDUKSIEPOTENSIALE

Increasing oxidising ability/Toenemende oksiderende vermoë

Half-reactions/Halfreaksies			E^{θ} (V)
$\text{Li}^+ + \text{e}^-$	$\rightleftharpoons$	Li	- 3,05
$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K	- 2,93
$\text{Cs}^+ + \text{e}^-$	$\rightleftharpoons$	Cs	- 2,92
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba	- 2,90
$\text{Sr}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sr	- 2,89
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca	- 2,87
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na	- 2,71
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg	- 2,36
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al	- 1,66
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn	- 1,18
$\text{Cr}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cr	- 0,91
$2\text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	$\text{H}_2(\text{g}) + 2\text{OH}^-$	- 0,83
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn	- 0,76
$\text{Cr}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Cr	- 0,74
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe	- 0,44
$\text{Cr}^{3+} + \text{e}^-$	$\rightleftharpoons$	Cr^{2+}	- 0,41
$\text{Cd}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cd	- 0,40
$\text{Co}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Co	- 0,28
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni	- 0,27
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn	- 0,14
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb	- 0,13
$\text{Fe}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Fe	- 0,06
$2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{H}_2(\text{g})$	0,00
$\text{S} + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{H}_2\text{S}(\text{g})$	+ 0,14
$\text{Sn}^{4+} + 2\text{e}^-$	$\rightleftharpoons$	Sn^{2+}	+ 0,15
$\text{Cu}^{2+} + \text{e}^-$	$\rightleftharpoons$	Cu^+	+ 0,16
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{g}) + 2\text{H}_2\text{O}$	+ 0,17
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu	+ 0,34
$2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$	$\rightleftharpoons$	4OH^-	+ 0,40
$\text{SO}_2 + 4\text{H}^+ + 4\text{e}^-$	$\rightleftharpoons$	$\text{S} + 2\text{H}_2\text{O}$	+ 0,45
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu	+ 0,52
$\text{I}_2 + 2\text{e}^-$	$\rightleftharpoons$	2I^-	+ 0,54
$\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O_2	+ 0,68
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	+ 0,77
$\text{NO}_3^- + 2\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{NO}_2(\text{g}) + \text{H}_2\text{O}$	+ 0,80
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag	+ 0,80
$\text{Hg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Hg}(\ell)$	+ 0,85
$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{NO}(\text{g}) + 2\text{H}_2\text{O}$	+ 0,96
$\text{Br}_2(\ell) + 2\text{e}^-$	$\rightleftharpoons$	2Br^-	+ 1,07
$\text{Pt}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pt	+ 1,20
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 2\text{H}_2\text{O}$	+ 1,23
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^-$	$\rightleftharpoons$	$2\text{H}_2\text{O}$	+ 1,23
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^-$	$\rightleftharpoons$	$2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+ 1,33
$\text{Cl}_2(\text{g}) + 2\text{e}^-$	$\rightleftharpoons$	2Cl^-	+ 1,36
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	+ 1,51
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$2\text{H}_2\text{O}$	+ 1,77
$\text{Co}^{3+} + \text{e}^-$	$\rightleftharpoons$	Co^{2+}	+ 1,81
$\text{F}_2(\text{g}) + 2\text{e}^-$	$\rightleftharpoons$	2F^-	+ 2,87

Increasing reducing ability/Toenemende reduserende vermoë